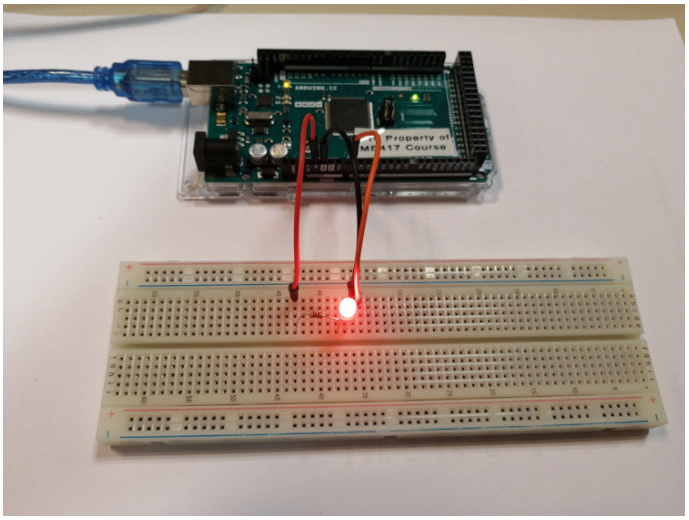


Figure 2: First step of Voltmeter with Arduino

As we can see from the Figure 2 shown above, +5V output from the Arduino is connected to positive side of the LED (where is the resistor connected to the LED) and GND is connected to the negative leg of the LED.

Normally, on the handout of the laboratory it is stated to measure voltage values between each leg of the resistor but voltage drop over the resistor can be calculated with Kirchhoff's Voltage Law, so we have measured voltage values of each leg of LED.

In order to get measurements we connected A0 analog input to negative leg of the LED the as shown in the Figure 2. With the analog input A0, we can read the data on Arduino.

To read the data coming from A0 analog input, we had to upload some codes to Arduino board because Since Arduino Mega's analog inputs have 10-bit resolution it maps 0-5V input voltage into integer values between 0 and 1023. So, the read input must be multiplied by $5/1023 = \approx 0.00488$ in order to have the real voltage value. As shown in the Figure 3 one we have implemented some codes to Arduino, verified the code and uploaded to Arduino as we have experienced in the laboratory 01. In the console we see the voltage values are measured as 0.00V in the red rectangle area.



Figure 3: Voltmeter code in Arduino and voltage values of first step

We will implement the same approach for the positive leg of the LED, we have connected the A0 analog input to the positive leg of the LED as shown in the Figure 4 and run the same code on Arduino to measure voltage value of the positive leg of LED.

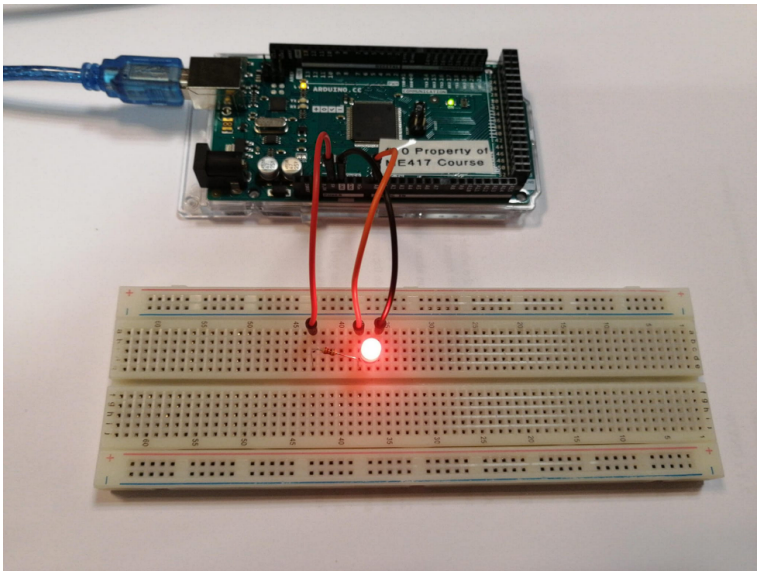


Figure 4: Second step of Voltmeter with Arduino

In the Figure 5 below here, we can see the measured value for the positive leg of the LED is measured as 2.01V.

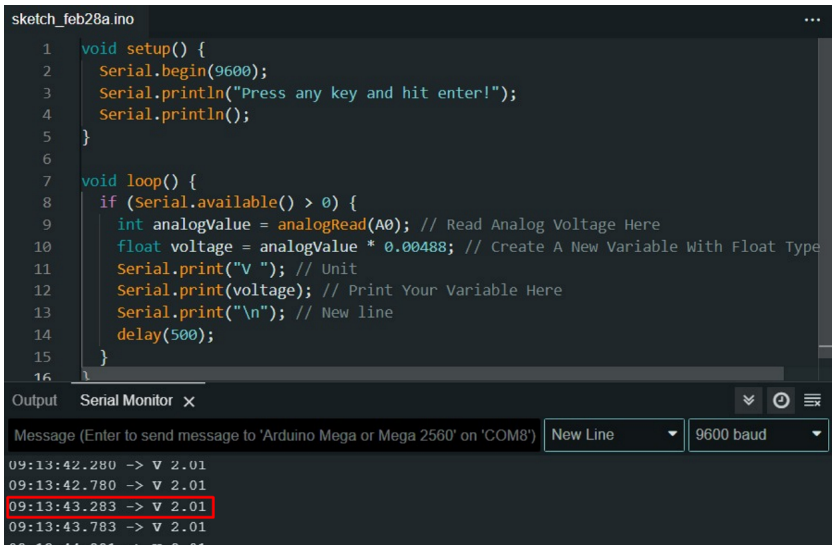


Figure 5: Voltmeter code in Arduino and voltage values of second step

We know the supplied voltage (V_S) is +5V and the voltage drop over the LED (V_{LED}) is measured alread as 2.01V, the voltage drop over the 220Ω resistor (V_R) can be calculated using Kirchhoff's Voltage Law.

$$V_S = V_{LED} + V_R -> 5V = 2.01V + V_R -> V_R = 2.99V$$

In the Figure 10 you can see the constructed circuit with CNY70. Red cable is +5V, black cable is GND and white cable reads output from the CNY70 sensor.

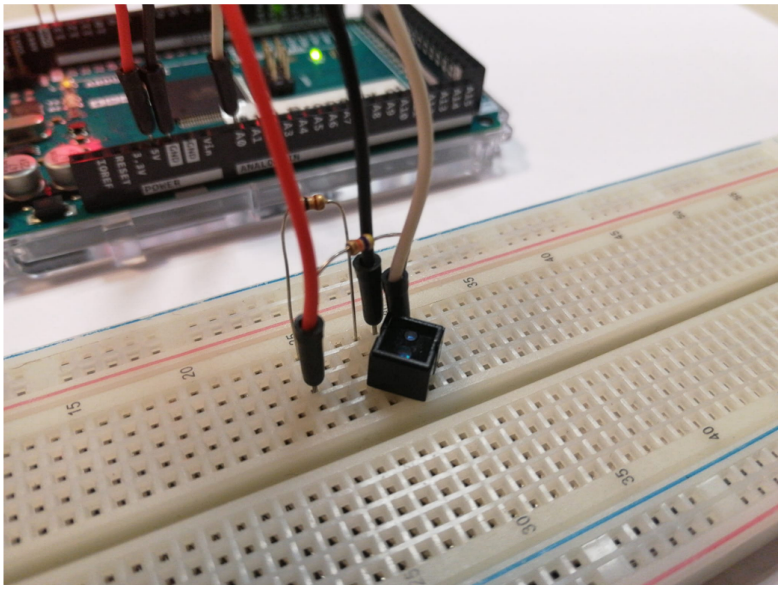


Figure 10: CNY70 sensor circuit

In order to operate the circuit we need an if-else loop implemented to the circuit. When the sensor voltage reading exceeds a pre-determined voltage level, the loop writes HIGH to digital pin 13 which is LED is connected to or setting the digital pin 13 output to LOW otherwise. By monitoring the voltage reading on the serial monitor while blocking and unblocking the sensor, that "certain value" was discovered.

sketch_feb28a.ino

```
1  void setup() {
2    Serial.begin(9600);
3    Serial.println("Press any key and hit enter!");
4    Serial.println();
5    pinMode(13, OUTPUT);
6  }
7
8  void loop() {
9    if (Serial.available() > 0) {
10     int analogValue = analogRead(A0); // Read Analog Voltage Here
11     float voltage = analogValue * 0.00488; // Create A New Variable With Float Type
12     Serial.print("V "); // Unit
13     Serial.print(voltage); // Print Your Variable Here
14     Serial.print("\n"); // New line
15     delay(200);
16     if (voltage > 0.2) {
17       digitalWrite(13, HIGH);
18     }
19     else{
20       digitalWrite(13, LOW);
21     }
22   }
23 }
```

Figure 11: CNY70 sensor circuit code implemented

As shown in the Figure 12, we can observe the circuit works and LED lights up while sensor is not detecting anything.

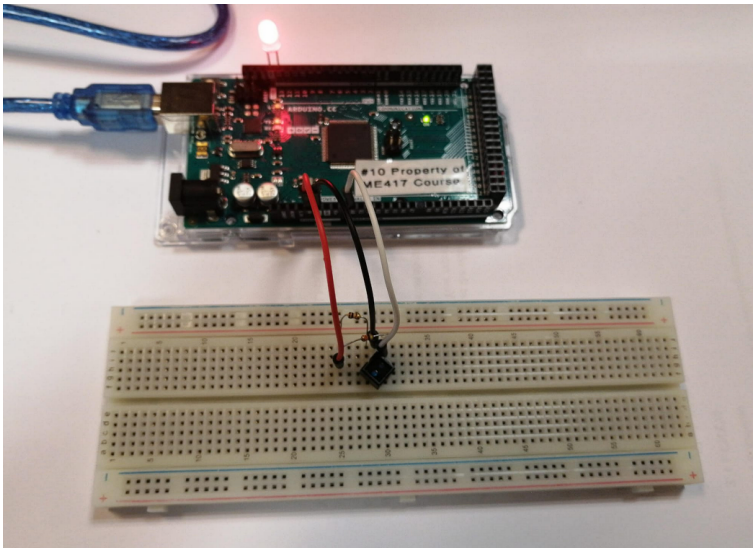


Figure 12: CNY70 sensor is not detecting anything and lights up

But when the sensor detects anything in dark colored (in our case it is dark brown colored pen) LED lights off as shown in the Figure 13.

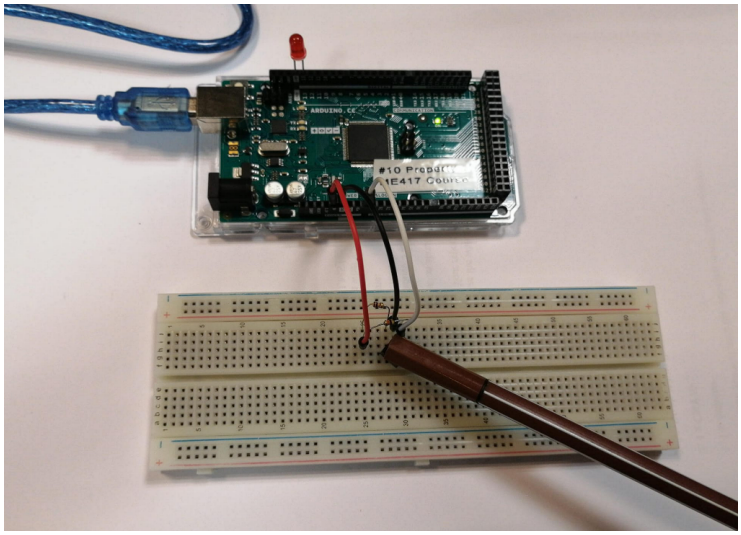


Figure 13: CNY70 sensor is detecting the pen and lights down